Candidate Name | Centre Number | Candidate Number

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS

PAPER 2

6032/2

1 hour 30 minutes

JUNE 2026 SESSION

Additional materials:
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

Answer **all** questions.

Write your answers in the spaces provided on the Question paper.

For numerical answers, **all** working should be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
6	
TOTAL	

This question paper consists of 17 printed pages.

DATA

speed of light in free space	$c = 3.00 \times 10^{8} \text{ms}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ Hm^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ Fm^{-1} $(1/4\pi\varepsilon_0 = 8.99 \times 10^{9}$ mF$^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}$ C
the Planck constant	$h = 6.63 \times 10^{-34}$ Js
unified atomic mass unit	$1\text{ u} = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ JK^{-1}mol^{-1}
the Avogadro constant	$N_A = 6.02 \times 10^{23}$mol^{-1}
the Boltzmann constant	$k = 1.38 \times 10^{-23}$ JK^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$Nm^{2}kg^{-2}
acceleration of free fall	$g = 9.81$ ms^{-2}

FORMULAE

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on/by a gas	$W = p\,\Delta V$
gravitational potential	$\phi = -Gm/r$
hydrostatic pressure	$p = \rho gh$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_o \cos \omega t$
	$v = \pm\omega\sqrt{(x_o^2 - x^2)}$
Attenuation of x-rays	$I = I_o e^{-\mu x}$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
capacitors in series	$1/C = 1/C_1 + 1/C_2 + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy of charged capacitor	$W = \frac{1}{2}QV$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_o \sin\omega t$
radioactive decay	$x = x_o \exp(-\lambda t)$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$

For Examiner's Use

1. (a) State the principle of *conservation of momentum*.

[1]

(b) A trolley of mass m_1, moving with a speed, u, makes a perfectly elastic collision with another trolley of mass, m_2, initially at rest. After the collision, trolley 2 moves off with $\frac{u}{3}$ speed along the same line.

Determine

1. the final velocity of m_1 in terms of u,

2. m_2 in terms of m_1.

[4]

For Examiner's Use

(c) State any **one** application of

(i) *Newton's first law of motion,*

....

.... [1]

(ii) *Newton's second law of motion.*

....

.... [1]

(d) A pellet of mass 3.5 g is fired vertically upwards into a block of moulding clay of mass 500 g. The pellet becomes embedded in the clay, and together they rise vertically through a distance of 6 cm.

(i) Calculate the gravitational potential energy gained by the clay and the pellet.

[1]

(ii) Show that the initial speed of the clay and the pellet after collision is $\approx 1.1\ ms^{-1}$.

[2]

For Examiner's Use

2. (a) (i) State any advantage of banking roads on curves.

....

.... [1]

(ii) Explain the term *free-fall.*

....

....

.... [2]

(b) **Fig. 2.1** shows a metal block of mass 1.5 kg suspended by a light inextensible string and completely immersed in water.

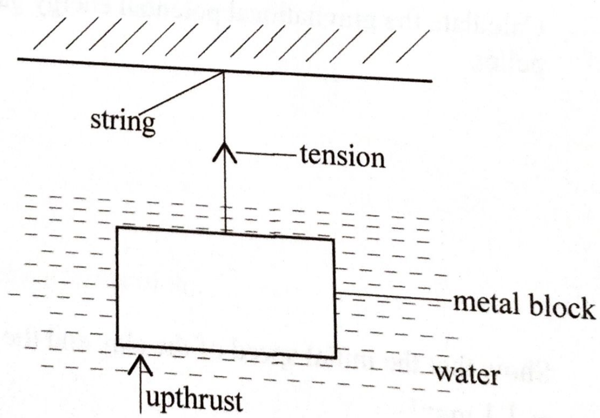

Fig .2.1

(i) Name the other force acting on the metal block.

.... [1]

(ii) The density of the block is $6.3 \times 10^3 \text{kgm}^{-3}$ and that of water is $1.0 \times 10^3 \text{ kgm}^{-3}$.

Determine the

1. upthrust acting on the block,

2. tension in the string.

[4]

For Examiner's Use

(c) A trolley of mass 3.2 kg is accelerated from rest by a driving force of 30 N to a speed of 18 ms^{-1} in 8 s.
Calculate the force of friction acting on the trolley.

[2]

3. (a) Explain the terms

(i) *free oscillations,*

....

.... [1]

(ii) *forced oscillations.*

....

.... [1]

(b) Give any **one** application of critical damping.

.... [1]

For Examiner's Use

(c) **Fig .3.1** shows a model for a CT scan. A section is divided into four voxels, with pixel **A, B, C,** and **D.**

For each rotation, the corresponding detector readings are indicated.

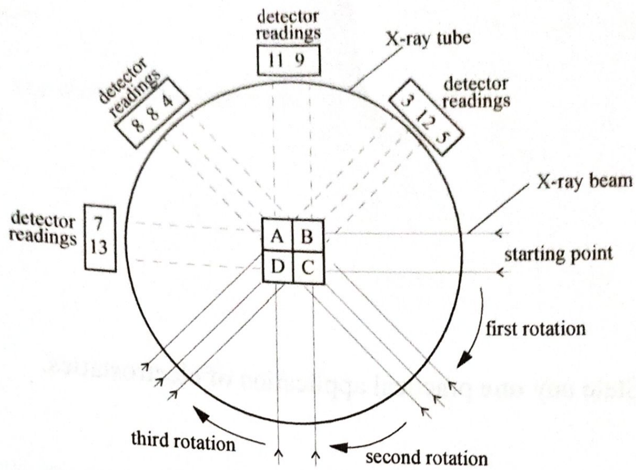

Fig .3.1

Determine the

(i) background count,

[1]

(ii) total voxel readings,

[3]

[Turn over

For Examiner's Use

(iii) pixel numbers **A, B, C** and **D**.

[3]

4. (a) State any **one** practical application of electrostatics.

.... [1]

(b) Draw the electric field pattern for a negative point charge.

[1]

(c) A charge of 2.6×10^{-15} C initially at rest, is moved through an electric field between two parallel plates separated by a distance of 3.5 cm. The potential difference between the plates is 8.0 kV.
Calculate

(i) the electric field strength,

[2]

(ii) the force acting on the charge.

[2]

(d) Name any **one** input transducer used in a recording studio.

.... [1]

(e) **Fig.4.1** shows a logic circuit with three inputs A, B and C.

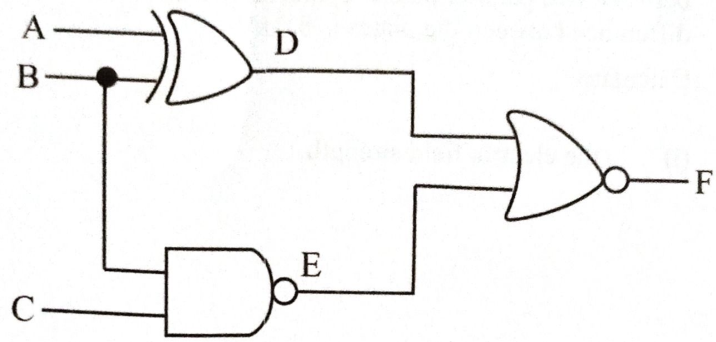

Fig.4.1

Construct the truth table for the logic circuit.

[3]

For Examiner's Use

5. (a) **Fig .5.1** shows the variation of extension, e, with tensile force, F, of a material.

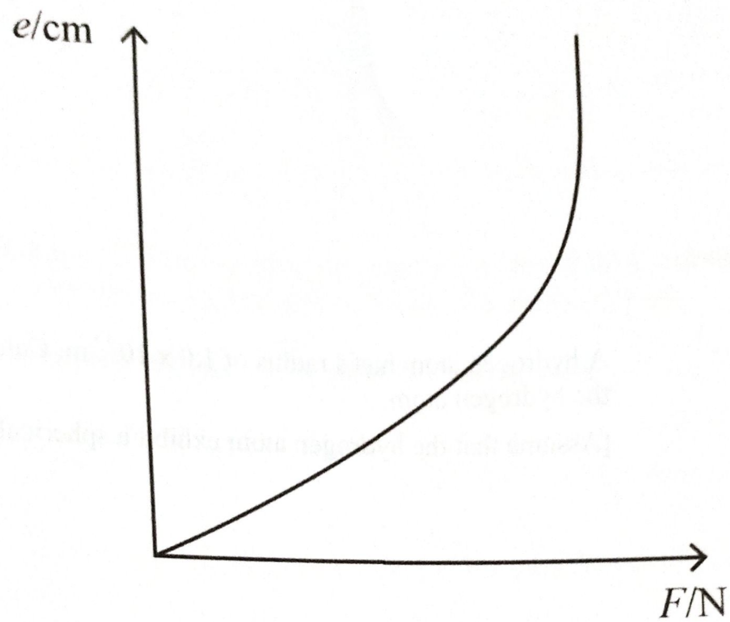

Fig .5.1

(i) State whether the material is ductile, brittle or polymeric.

....

.... [1]

(ii) On **Fig 5.1**, indicate with the letter P, the limit of proportionality. [1]

(iii) Explain how the spring constant of the material may be obtained from **Fig 5.1**.

....

....

.... [2]

For Examiner's Use

(b) Four hydrogen molecules have velocities 300 ms^{-1}, 380 ms^{-1}, 400 ms^{-1} and 420 ms^{-1}.

Calculate their *root-mean-square (r.m.s) speed.*

[2]

(c) A hydrogen atom has a radius of 1.0×10^{-15} m. Calculate the density of the hydrogen atom.

[Assume that the hydrogen atom exhibit a spherical proton]

[2]

(d) "When ice blocks at 0 °C are heated with a constant heat supply, their temperature does not change."

(i) State the physical change that occurs to the ice blocks.

....[1]

For Examiner's Use

(ii) Explain why the temperature of the ice blocks does not change.

....

....

.... [1]

6. (a) In Millikan's oil drop experiment, a charged oil drop falls at constant speed when there is no potential difference, between the plates.

(i) Explain why the oil drop falls at constant speed.

....

....

.... [1]

(ii) Another oil drop of mass 4.0×10^{-50} kg carrying a charge of 9.6×10^{-19} C is held stationary by an electric field.

Calculate the electric field strength.

[2]

[Turn over

For Examiner's Use

(b) An LED provides an input power of 2.12 mW to an optic fibre of length 60 m. The output power at the other end of the optic fibre is 2.05 mW.

Calculate the attenuation per unit length.

[2]

(c) **(i)** Explain the meaning of the term *excitation energy*.

....

....

....

[1]

(ii) Calculate the minimum speed of an incident electron required to excite an atom from its ground state energy level of $-13.6\ eV$ to an excited state of $-10.2\ eV$.

[3]

For Examiner's Use

(d) Explain the meaning of the term *modulation*.

....

....

....

[1]

Blank Page